

PRACTICAL 5

1. Title: Apply array methods and object handling; create a cart total calculator with discount logic.

2. Software/Tools Required: Visual Studio Code (VS Code), JavaScript.

3. Theory:

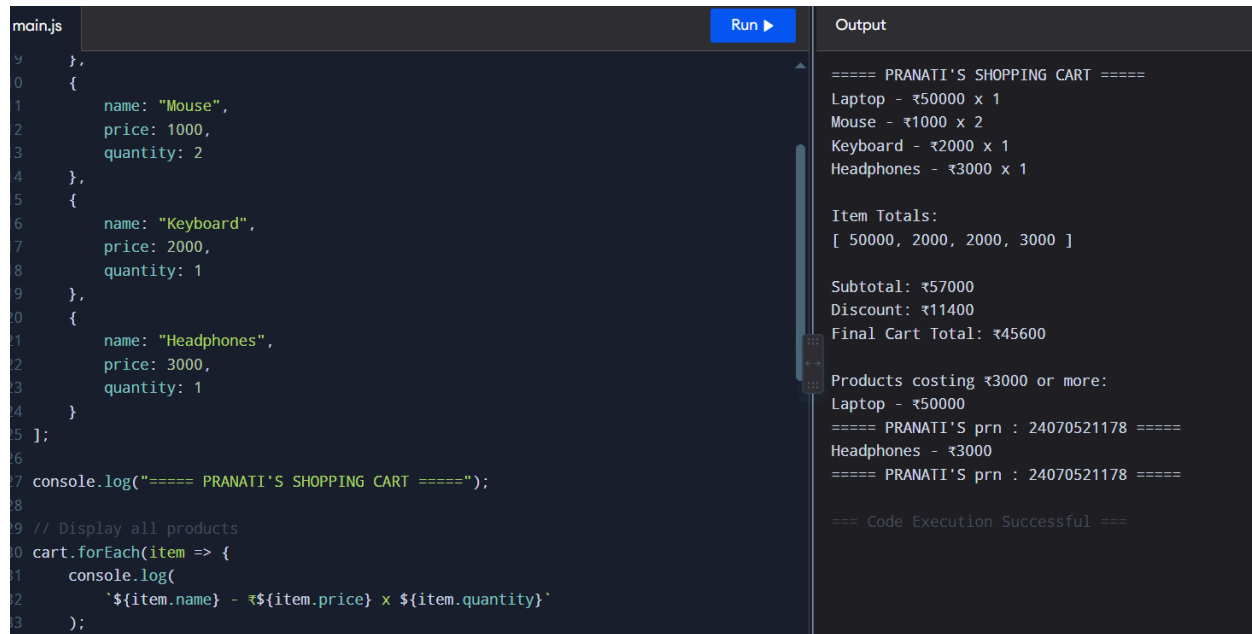
JavaScript objects allow storing collections of data using key-value pairs (e.g., name, price, quantity), while arrays store ordered lists of these objects. Higher-order array methods

such as `reduce()` process elements systematically without standard explicit loops. The `reduce()` method executes a reducer function on each array element, accumulating values to

compute a single output—such as the total cost of all products.

Conditional statements (if-else or ternary operators) evaluate business logic, such as applying a 10% discount if the cart's total exceeds a specific threshold.

4. Output:



```
main.js Run ▶  
9 },  
10 {  
11   name: "Mouse",  
12   price: 1000,  
13   quantity: 2  
14 },  
15 {  
16   name: "Keyboard",  
17   price: 2000,  
18   quantity: 1  
19 },  
20 {  
21   name: "Headphones",  
22   price: 3000,  
23   quantity: 1  
24 }  
25 ];  
26  
27 console.log("==== PRANATI'S SHOPPING CART ====");  
28  
29 // Display all products  
30 cart.forEach(item => {  
31   console.log(  
32     `${item.name} - ₹${item.price} x ${item.quantity}`  
33   );  
34 });
```

Output

```
==== PRANATI'S SHOPPING CART ====  
Laptop - ₹50000 x 1  
Mouse - ₹1000 x 2  
Keyboard - ₹2000 x 1  
Headphones - ₹3000 x 1  
  
Item Totals:  
[ 50000, 2000, 2000, 3000 ]  
  
Subtotal: ₹57000  
Discount: ₹11400  
Final Cart Total: ₹45600  
  
Products costing ₹3000 or more:  
Laptop - ₹50000  
==== PRANATI'S prn : 24070521178 ====  
Headphones - ₹3000  
==== PRANATI'S prn : 24070521178 ====  
  
=== Code Execution Successful ===
```

5. Result/Conclusion: The experiment was performed successfully. JavaScript arrays, objects, and higher-order array methods (specifically `reduce()`) were demonstrated to process structured data efficiently. Conditional logic for applying discounts was implemented, and the final output was verified using browser `console.log()` methods.



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Case Study : 5

1. Title: Study of Arrays and Loops in JavaScript with Maximum and Minimum Value Calculation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Arrays in JavaScript are used to store multiple values in a single variable. Array methods such as `push()`, `pop()`, `shift()`, `unshift()`, `splice()`, and `slice()` are used to add, remove, and extract elements from an array. Methods like `map()`, `filter()`, `reduce()`, and `forEach()` help in processing and manipulating array elements. In this case study, an array of numbers is created, and JavaScript is used to find the maximum and minimum values using `Math.max()` and `Math.min()`.

4. Output:

```
Arrays and Loops in JavaScript

Original Array: 10,20,30,40,50

After push(60): 10,20,30,40,50,60
After pop(): 10,20,30,40,50
After unshift(5): 5,10,20,30,40,50
After shift(): 10,20,30,40,50
After splice(): 10,20,35,40,50
After slice(1,4): 20,35,40

forEach(): 10 20 35 40 50
map() - Doubled Array: 20,40,70,80,100
filter() - Numbers > 20: 35,40,50
reduce() - Sum: 155

Case Study
Array: 10,20,35,40,50
Maximum Value: 50
Minimum Value: 10
```



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Conclusion : In this experiment, we successfully studied arrays and loops in JavaScript. We implemented different array methods to add, remove, and manipulate elements. We also used `map()`, `filter()`, `reduce()`, and `forEach()` for array processing. The case study helped us find the maximum and minimum values from an array of numbers and understand the practical use of JavaScript array methods.